

APPENDIX PROOF OF PROPOSITION 1

Let $\Omega = \text{conv}\{\mathbf{c}_1, \dots, \mathbf{c}_M\}$, where $M \geq 3$ distinct receivers satisfy

$$\mathbf{c}_m \notin \text{conv}\{\mathbf{c}_j : j \in [M], j \neq m\}, \quad m \in [M].$$

Assume that $\mathbf{p}^\dagger \in \text{int } \Omega$ and $h_m \neq 0$ for every m . We consider the noiseless continuous-frequency model with $0 < P_- \leq P(f) \leq P_+ < \infty$ on $[-B/2, B/2]$, and use $\kappa = P_+/P_-$ and $D = \max_{i < j} \|\mathbf{c}_i - \mathbf{c}_j\|_2$ as in Proposition 1. Position gradients are evaluated on $\Omega \setminus \{\mathbf{c}_1, \dots, \mathbf{c}_M\}$.

We first establish a positive lower bound on the gradient of the squared TDOA mismatch V away from $\mathbf{p}^\dagger$. We then show that Gaussian smoothing makes the scaled DPD gradient approximate $\nabla_{\mathbf{p}} V$ closely enough to exclude other stationary points.

A.1. Geometric gradient bound

To express V in terms of the relative received powers, write

$$S = \sum_{m=1}^M |h_m|^2, \quad q_m = \frac{|h_m|^2}{S}, \quad a_m = \sqrt{q_m}. \quad (\text{A.1})$$

Then $\sum_m q_m = \|\mathbf{a}\|_2^2 = 1$ and

$$V(\mathbf{p}) = c^2 \sum_{i < j} q_i q_j \Delta_{ij}^2(\mathbf{p}). \quad (\text{A.2})$$

The geometric bound needed for the gradient comparison is

$$\mu_{\text{geo}} = \inf_{\substack{\mathbf{p} \in \Omega \setminus \{\mathbf{c}_1, \dots, \mathbf{c}_M\} \\ \mathbf{p} \neq \mathbf{p}^\dagger}} \frac{\|\nabla_{\mathbf{p}} V(\mathbf{p})\|_2}{\|\mathbf{p} - \mathbf{p}^\dagger\|_2} > 0. \quad (\text{A.3})$$

The smoothing threshold will be obtained with

$$C_{\text{geo}} = \frac{36\pi^2}{\mu_{\text{geo}}}. \quad (\text{A.4})$$

For a fixed trial position, define its range errors and the unit directions from the receivers by

$$e_m = \|\mathbf{p} - \mathbf{c}_m\|_2 - \|\mathbf{p}^\dagger - \mathbf{c}_m\|_2, \quad \mathbf{u}_m = \frac{\mathbf{p} - \mathbf{c}_m}{\|\mathbf{p} - \mathbf{c}_m\|_2},$$

and let $\bar{e} = \sum_m q_m e_m$. Expanding the pairwise differences in (A.2) gives

$$V(\mathbf{p}) = \sum_m q_m (e_m - \bar{e})^2, \quad (\text{A.5})$$

$$\frac{1}{2} \nabla_{\mathbf{p}} V(\mathbf{p}) = \sum_m q_m (e_m - \bar{e}) \mathbf{u}_m. \quad (\text{A.6})$$

We therefore need to show that the terms in (A.6) cannot cancel when $\mathbf{p} \neq \mathbf{p}^\dagger$.

The equality $V(\mathbf{p}) = 0$ means that $\mathbf{p}$ and $\mathbf{p}^\dagger$ produce identical TDOAs [1]. Suppose that this holds for $\mathbf{p} \neq \mathbf{p}^\dagger$. Then all range errors equal one number. If that number were nonnegative, every receiver would lie in the closed half-plane of points at least as close to $\mathbf{p}^\dagger$ as to $\mathbf{p}$. Their convex hull

could not contain $\mathbf{p}$. If it were negative, the reverse half-plane would exclude $\mathbf{p}^\dagger$. Both contradict the positions belonging to Ω . Hence $V(\mathbf{p}) > 0$ for every $\mathbf{p} \neq \mathbf{p}^\dagger$ in Ω .

To establish the ordering of the range errors along the polygon, take $\mathbf{p} \in \text{int } \Omega$, $\mathbf{p} \neq \mathbf{p}^\dagger$, and consider, along its boundary, the function

$$E(\mathbf{z}) = \|\mathbf{z} - \mathbf{p}\|_2 - \|\mathbf{z} - \mathbf{p}^\dagger\|_2.$$

A supporting line of a convex polygon places all its interior points on the same side [2, Sec. 2.5]. This prevents E from having an extremum along an edge where $\nabla E \neq 0$. At such a point, set $\mathbf{u} = (\mathbf{z} - \mathbf{p})/\|\mathbf{z} - \mathbf{p}\|_2$ and $\mathbf{v} = (\mathbf{z} - \mathbf{p}^\dagger)/\|\mathbf{z} - \mathbf{p}^\dagger\|_2$. Then

$$\nabla E(\mathbf{z})^\top (\mathbf{p} - \mathbf{z}) = -\|\mathbf{z} - \mathbf{p}\|_2 (1 - \mathbf{u}^\top \mathbf{v}) < 0,$$

$$\nabla E(\mathbf{z})^\top (\mathbf{p}^\dagger - \mathbf{z}) = \|\mathbf{z} - \mathbf{p}^\dagger\|_2 (1 - \mathbf{u}^\top \mathbf{v}) > 0.$$

The line normal to $\nabla E(\mathbf{z})$ therefore separates the two interior points, whereas a supporting line cannot do so. At a polygon vertex, a local extremum along the boundary would require the directional derivatives along the two incident edges to have the same sign. One of ∇E and $-\nabla E$ would then define a supporting line at that vertex, which gives the same contradiction.

Consequently, along the polygon boundary, E has extrema only where the two unit directions coincide. These are the intersections with the two rays extending the segment between $\mathbf{p}$ and $\mathbf{p}^\dagger$, where E equals $\pm\|\mathbf{p} - \mathbf{p}^\dagger\|_2$. On each of the two boundary arcs joining these points, E is strictly monotone. It follows that the positive and negative values of $e_m - \bar{e}$ occupy two separate consecutive groups of vertices. Both groups are nonempty because $V(\mathbf{p}) > 0$.

Viewed from an interior $\mathbf{p}$, the directions $\mathbf{u}_m$ have the same cyclic order as the vertices. Their two sign groups occupy disjoint arcs of the unit circle. The convex hulls of these groups are disjoint and can be strictly separated [2, Sec. 2.5.1]. Consequently, there exist $\mathbf{v} \in \mathbb{R}^2$ and $b \in \mathbb{R}$ such that $(e_m - \bar{e})(\mathbf{v}^\top \mathbf{u}_m - b) > 0$ whenever $e_m \neq \bar{e}$. Since $\sum_m q_m (e_m - \bar{e}) = 0$, (A.6) yields

$$\frac{1}{2} \mathbf{v}^\top \nabla_{\mathbf{p}} V = \sum_m q_m (e_m - \bar{e})(\mathbf{v}^\top \mathbf{u}_m - b) > 0. \quad (\text{A.7})$$

Hence V has no stationary point other than $\mathbf{p}^\dagger$ in the polygon interior.

The separation also persists in limits to the polygon boundary. At an edge point that is not a receiver, the receiver directions remain distinct and retain their cyclic order. At a receiver vertex $\mathbf{c}_j$, the possible limits of $\mathbf{u}_j$ lie in the inward tangent cone, while all other $\mathbf{u}_m$ lie in its negative cone. These cones have angle less than π , so their unit directions do not coincide. Moreover, e_j is strictly smaller than every other e_m , and therefore $e_j - \bar{e} < 0$. For each such limit the two sign groups remain separable, and (A.7) is still strict. Compactness of the boundary and of the limiting directions

then gives a positive lower bound on $\|\nabla V\|_2$ outside any neighborhood of $\mathbf{p}^\natural$.

Near $\mathbf{p}^\natural$, the required lower bound follows from

$$\nabla_{\mathbf{p}}^2 V(\mathbf{p}^\natural) = 2 \sum_m q_m (\mathbf{u}_m - \bar{\mathbf{u}})(\mathbf{u}_m - \bar{\mathbf{u}})^\top, \quad (\text{A.8})$$

where $\bar{\mathbf{u}} = \sum_m q_m \mathbf{u}_m$ and the directions are evaluated at $\mathbf{p}^\natural$. For any nonzero $\mathbf{v}$, the corresponding quadratic form is $2 \sum_m q_m [\mathbf{v}^\top (\mathbf{u}_m - \bar{\mathbf{u}})]^2$. It can vanish only if all $\mathbf{u}_m$ lie on one affine line. Three distinct points on the unit circle cannot satisfy this condition, so the Hessian is positive definite. Together with the preceding separation, this proves (A.3) for the stated polygon geometry. Finally, the trace of (A.8) is at most 2. Since this Hessian has two positive eigenvalues, its smaller eigenvalue is at most 1. Taking the limit in (A.3) along an associated eigenvector gives

$$0 < \mu_{\text{geo}} \leq 1. \quad (\text{A.9})$$

A.2. Gaussian-weighted spectral moments

To bound the frequency moments in the gradient approximation, write the Gaussian-weighted matrix as

$$[\mathbf{Q}_\alpha(\mathbf{p})]_{ij} = h_i h_j^* \int_{-B/2}^{B/2} P(f) e^{-\alpha(2f/B)^2} e^{j2\pi f \Delta_{ij}(\mathbf{p})} df. \quad (\text{A.10})$$

Let $Z_\alpha = \int_{-B/2}^{B/2} P(f) e^{-\alpha(2f/B)^2} df$ and introduce the normalized spectral density

$$\pi_\alpha(f) = Z_\alpha^{-1} P(f) e^{-\alpha(2f/B)^2}.$$

For f distributed according to π_α , put

$$\begin{aligned} \bar{f}_\alpha &= \mathbb{E}f, & X &= 2\pi(f - \bar{f}_\alpha), \\ m_{r,\alpha} &= \mathbb{E}X^r, & r &= 2, 4. \end{aligned} \quad (\text{A.11})$$

The ratio $\eta_\alpha = m_{4,\alpha}/m_{2,\alpha}$ will control the terms beyond the quadratic mismatch in the DPD gradient. We show that

$$0 < m_{2,\alpha} \leq \eta_\alpha \leq \frac{3\pi^2 \kappa^2 B^2}{2\alpha}. \quad (\text{A.12})$$

The first inequality holds because P is positive on an interval. The second follows from $m_{2,\alpha}^2 \leq m_{4,\alpha}$.

To relate the weighted spectrum to a Gaussian density, set $x = 2\pi f$ and $\sigma_\alpha^2 = \pi^2 B^2 / (2\alpha)$. Let G be the centered Gaussian density proportional to $e^{-x^2/(2\sigma_\alpha^2)}$ on $[-\pi B, \pi B]$, and write $b_r = \mathbb{E}_G x^r$. Integration by parts, retaining the nonnegative boundary terms, gives

$$b_2 \leq \sigma_\alpha^2, \quad b_4 \leq 3\sigma_\alpha^2 b_2. \quad (\text{A.13})$$

Relative to G , the spectral density is

$$R(x) = \frac{P(x/(2\pi))}{\mathbb{E}_G[P(x/(2\pi))]}.$$

It satisfies $\mathbb{E}_G R = 1$ and $R \leq \kappa$. The actual spectrum may have a nonzero mean. Writing $m = \mathbb{E}_G[Rx] = 2\pi \bar{f}_\alpha$, the

Cauchy–Schwarz inequality gives

$$m^2 = (\mathbb{E}_G[(R-1)x])^2 \leq (\mathbb{E}_G R^2 - 1)b_2 \leq (\kappa - 1)b_2. \quad (\text{A.14})$$

Using the ratio P_+/P_- in the centered moment integrals, then (A.13) and (A.14), we obtain

$$\begin{aligned} \eta_\alpha &\leq \kappa \frac{\mathbb{E}_G(x-m)^4}{\mathbb{E}_G(x-m)^2} = \kappa \frac{b_4 + 6m^2 b_2 + m^4}{b_2 + m^2} \\ &\leq 3\kappa(\sigma_\alpha^2 + m^2) \leq 3\kappa^2 \sigma_\alpha^2. \end{aligned} \quad (\text{A.15})$$

This proves (A.12).

A.3. Matrix representation using centered frequencies

We remove the mean-frequency phase to use the centered moments in (A.11). Define

$$[\mathbf{A}_\alpha(\mathbf{p})]_{ij} = a_i a_j \mathbb{E}[e^{jX \Delta_{ij}(\mathbf{p})}]. \quad (\text{A.16})$$

Define a diagonal unitary matrix with entries

$$[\mathbf{U}_\alpha(\mathbf{p})]_{mm} = e^{j \arg h_m} e^{j2\pi \bar{f}_\alpha [\tau_m(\mathbf{p}) - \tau_m(\mathbf{p}^\natural)]}.$$

Then

$$\frac{\mathbf{Q}_\alpha(\mathbf{p})}{SZ_\alpha} = \mathbf{U}_\alpha(\mathbf{p}) \mathbf{A}_\alpha(\mathbf{p}) \mathbf{U}_\alpha^\text{H}(\mathbf{p}). \quad (\text{A.17})$$

Unitary similarity preserves eigenvalues [3, Sec. 2.2]. Since $\text{tr}(\mathbf{Q}_\alpha) = SZ_\alpha$, the normalized DPD objective is $\ell_\alpha = \lambda_{\max}(\mathbf{A}_\alpha)$. At $\mathbf{p}^\natural$, every Δ_{ij} is zero, and hence $\mathbf{A}_\alpha = \mathbf{a}\mathbf{a}^\top$ and $\ell_\alpha = 1$.

We next bound the departure from this rank-one matrix, together with its position derivative. In the following estimates, $\partial_{\mathbf{v}}$ denotes differentiation in a real unit direction $\mathbf{v}$, and $r = \|\mathbf{p} - \mathbf{p}^\natural\|_2$. The triangle inequality and the delay derivatives give

$$|\Delta_{ij}| \leq \frac{2 \min\{D, r\}}{c}, \quad |\partial_{\mathbf{v}} \Delta_{ij}| \leq \frac{2}{c}. \quad (\text{A.18})$$

As $\mathbb{E}X = 0$, the elementary inequalities $|e^{jt} - 1 - jt| \leq t^2/2$ and $|e^{jt} - 1| \leq |t|$ imply

$$\|\mathbf{A}_\alpha - \mathbf{a}\mathbf{a}^\top\|_2 \leq 2m_{2,\alpha}(D/c)^2 =: \varepsilon, \quad (\text{A.19})$$

$$\|\partial_{\mathbf{v}} \mathbf{A}_\alpha\|_2 \leq \frac{4m_{2,\alpha} r}{c^2}. \quad (\text{A.20})$$

These bounds follow by first estimating the individual entries,

$$|[\mathbf{A}_\alpha - \mathbf{a}\mathbf{a}^\top]_{ij}| \leq a_i a_j m_{2,\alpha} \Delta_{ij}^2/2.$$

and

$$|\partial_{\mathbf{v}} \mathbf{A}_\alpha|_{ij}| \leq a_i a_j m_{2,\alpha} |\Delta_{ij}| |\partial_{\mathbf{v}} \Delta_{ij}|.$$

The second inequality uses $\mathbb{E}[jX e^{jX \Delta}] = \mathbb{E}[jX(e^{jX \Delta} - 1)]$. Taking Frobenius norms and using $\sum_{i,j} a_i^2 a_j^2 = 1$ gives (A.19) and (A.20).

A.4. Gradient approximation error

The matrix bound controls both the leading eigenvalue and its eigenvector, which are needed to differentiate ℓ_α . Let $\varepsilon \leq 1/4$. By the Hermitian eigenvalue perturbation inequality in [3, Thm. 4.3.1], the largest eigenvalue of $\mathbf{A}_\alpha$ is at least $1 - \varepsilon$, whereas all the others are at most ε . The largest eigenvalue is therefore simple.

The eigenvector perturbation is controlled by the eigenvalue gap [4]. Let $\boldsymbol{\nu}$ be a corresponding unit eigenvector. Projecting its eigenvalue equation onto the orthogonal complement of $\mathbf{a}$ and using $\|\mathbf{A}_\alpha - \mathbf{a}\mathbf{a}^\top\|_2 \leq \varepsilon$ gives

$$\|(\mathbf{I} - \mathbf{a}\mathbf{a}^\top)\boldsymbol{\nu}\|_2 \leq \frac{\varepsilon}{1 - \varepsilon}. \quad (\text{A.21})$$

The derivative of a simple eigenvalue is given by [5, Thm. 1.1]

$$\partial_{\mathbf{v}} \ell_\alpha = \boldsymbol{\nu}^H (\partial_{\mathbf{v}} \mathbf{A}_\alpha) \boldsymbol{\nu}.$$

To replace $\boldsymbol{\nu}$ by $\mathbf{a}$, note that the projector difference $\boldsymbol{\nu}\boldsymbol{\nu}^H - \mathbf{a}\mathbf{a}^\top$ has eigenvalues $\pm\|(\mathbf{I} - \mathbf{a}\mathbf{a}^\top)\boldsymbol{\nu}\|_2$ and zeros. Thus the difference between the two quadratic forms is bounded by twice the matrix norm times $\|(\mathbf{I} - \mathbf{a}\mathbf{a}^\top)\boldsymbol{\nu}\|_2$. Together with (A.20) and (A.21), this gives

$$|\partial_{\mathbf{v}} \ell_\alpha - \mathbf{a}^\top (\partial_{\mathbf{v}} \mathbf{A}_\alpha) \mathbf{a}| \leq \frac{16m_{2,\alpha}^2 D^2}{1 - \varepsilon} r. \quad (\text{A.22})$$

To relate this quadratic form to $\nabla_{\mathbf{p}} V$, pair the conjugate entries in (A.16) to obtain

$$\mathbf{a}^\top (\partial_{\mathbf{v}} \mathbf{A}_\alpha) \mathbf{a} = - \sum_{i,j} q_i q_j \mathbb{E}[X \sin(X \Delta_{ij})] \partial_{\mathbf{v}} \Delta_{ij}.$$

Because $\sum_{i,j} q_i q_j \Delta_{ij} \partial_{\mathbf{v}} \Delta_{ij} = \partial_{\mathbf{v}} V / c^2$, the bound $|\sin t - t| \leq |t|^3/6$, together with (A.18), shows that

$$\left| \mathbf{a}^\top (\partial_{\mathbf{v}} \mathbf{A}_\alpha) \mathbf{a} + \frac{m_{2,\alpha}}{c^2} \partial_{\mathbf{v}} V \right| \leq \frac{8m_{4,\alpha}}{3} \frac{D^2}{c^4} r. \quad (\text{A.23})$$

Combining (A.22) and (A.23), and using $m_{2,\alpha}^2 \leq m_{4,\alpha}$, we have $16/(1 - \varepsilon) + 8/3 \leq 24$. Since the estimate holds for every unit $\mathbf{v}$, it follows that

$$\left\| \nabla_{\mathbf{p}} \frac{c^2(1 - \ell_\alpha)}{m_{2,\alpha}} - \nabla_{\mathbf{p}} V \right\|_2 \leq 24\eta_\alpha \left(\frac{D}{c} \right)^2 \|\mathbf{p} - \mathbf{p}^\natural\|_2. \quad (\text{A.24})$$

A.5. Smoothing threshold and uniqueness

To keep the gradient approximation error below the geometric lower bound, choose

$$\alpha > \frac{36\pi^2}{\mu_{\text{geo}}} \kappa^2 \left(\frac{BD}{c} \right)^2. \quad (\text{A.25})$$

By (A.12), $24\eta_\alpha (D/c)^2 < \mu_{\text{geo}}$. In particular, (A.9) gives $\varepsilon = 2m_{2,\alpha} (D/c)^2 < \mu_{\text{geo}}/12 \leq 1/12$, so the eigenvalue and

gradient estimates above apply. Using (A.3) and the reverse triangle inequality in (A.24), we obtain

$$\left\| \nabla_{\mathbf{p}} \frac{c^2(1 - \ell_\alpha(\mathbf{p}))}{m_{2,\alpha}} \right\|_2 \geq \left[\mu_{\text{geo}} - 24\eta_\alpha \left(\frac{D}{c} \right)^2 \right] \|\mathbf{p} - \mathbf{p}^\natural\|_2 > 0 \quad (\mathbf{p} \neq \mathbf{p}^\natural). \quad (\text{A.26})$$

The scaling factors are positive and independent of $\mathbf{p}$, so (A.26) excludes false stationary points of ℓ_α and $\lambda_{\max}(\mathbf{Q}_\alpha)$. At $\mathbf{p}^\natural$, (A.16) and $\mathbb{E}X = 0$ give $\nabla_{\mathbf{p}} \ell_\alpha = 0$.

Moreover, $\mathbf{A}_\alpha \succeq 0$ and $\text{tr}(\mathbf{A}_\alpha) = 1$ imply $\ell_\alpha \leq 1$, with equality at $\mathbf{p}^\natural$. Equality requires $\mathbf{A}_\alpha$ to have rank one, so $e^{j2\pi f \Delta_{ij}}$ is constant on the frequency interval. Since $P(f) > 0$ there, $\Delta_{ij} = 0$ for all $i < j$. Section A.1 then gives $\mathbf{p} = \mathbf{p}^\natural$. This proves Proposition 1 with $C_{\text{geo}} = 36\pi^2/\mu_{\text{geo}}$. $\square$

A.6. Finite-frequency objective

For the finite sum in (7), define

$$\pi_{k,\alpha} = \frac{\omega_k(\alpha) P_k}{\sum_j \omega_j(\alpha) P_j}, \quad \bar{f}_\alpha = \sum_k \pi_{k,\alpha} f_k,$$

and $m_{r,\alpha} = \sum_k \pi_{k,\alpha} [2\pi(f_k - \bar{f}_\alpha)]^r$ for $r = 2, 4$. With expectations taken over these weighted frequencies, the matrix representation and gradient bounds in Sections A.3 and A.4 apply directly. The resulting sufficient condition is

$$m_{2,\alpha} > 0, \quad 24 \frac{m_{4,\alpha}}{m_{2,\alpha}} \left(\frac{D}{c} \right)^2 < \mu_{\text{geo}}. \quad (\text{A.27})$$

This condition makes (A.26) strictly positive away from $\mathbf{p}^\natural$, establishing the same unique-stationary-point result for the finite-frequency objective on $\Omega \setminus \{\mathbf{c}_1, \dots, \mathbf{c}_M\}$.

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